RouteRec: Behavior-Guided Expert Routing for Sequential Recommendation

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Abstract

Sequential recommendation is commonly optimized with one shared computation pipeline, even though interaction regimes vary substantially across users, sessions, and time scales. In practice, some prefixes are dominated by stable preference, some by repetition, and others by local exploration. This mismatch between behavioral diversity and shared computation is increasingly visible as backbones become stronger: performance improves, but the model still offers little control over *how* different behavior patterns should trigger different computation.

Mixture-of-experts (MoE) is a natural candidate for conditional computation, yet many sequential MoE variants still route primarily from hidden states. This creates a semantic bottleneck: the same latent signal is asked to encode sequence content and decide routing allocation. As a result, expert assignment is often effective but opaque, making it difficult to interpret specialization, diagnose failure modes, or align routing with observable behavioral factors.

We propose **RouteRec**, a behavior-guided MoE framework that explicitly decouples routing signals from representation signals. RouteRec keeps a strong sequence encoder for hidden-state construction, while routing is driven by compact behavior cues extracted from standard logs (timestamps, optional categories, and interaction statistics). These cues are organized into four semantic families (Tempo, Focus, Memory, Exposure) and consumed through three temporal scopes (macro, mid, micro), so the routing structure matches evidence granularity rather than imposing a single gating bias everywhere.

This design is intentionally lightweight: we avoid heavy side-information modeling in the routing branch and instead emphasize interpretable, transferable summaries that are easy to compute across datasets. We further provide a dedicated preliminary formulation to make the method readable without overloading notation. Experiments show that RouteRec improves next-item recommendation over strong baselines, and diagnostic analyses indicate that improvements are associated with more behavior-consistent expert usage rather than unstructured parameter scaling.

CCS Concepts

• **Information systems** → **Recommender systems**.

Keywords

sequential recommendation, mixture of experts, conditional computation, routing, behavioral features

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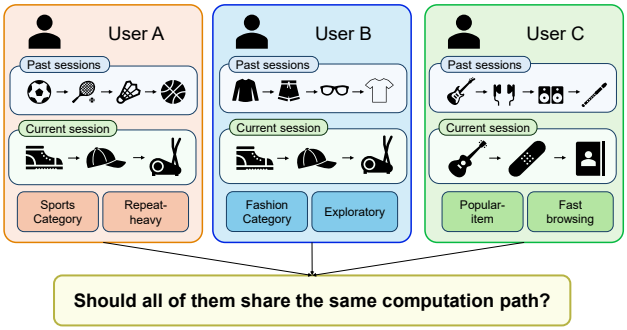

Figure 1: Different users and sessions can require different computational paths.

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1 Introduction

Sequential recommendation is usually formulated as next-item prediction over an ordered interaction history. At first glance, this appears to be a homogeneous prediction task, but real user behavior is not homogeneous at all. Even for the same user, one session may reflect persistent preference, another may reflect short-term mission behavior, and another may be dominated by quick exploratory clicks. Prior surveys and studies consistently describe this multi-regime nature of sequential and session-based recommendation [24, 25].

Modern backbones have become very strong at sequence modeling [5, 8, 11, 14, 17, 19], but they still predominantly rely on shared transformations across behaviorally different cases. This design is efficient and often performant on average, yet it leaves an important modeling question open: when two prefixes imply fundamentally different behavioral contexts, should they really invoke the same computation path?

Figure 1 provides an intuitive view of this gap. Distinct users and sessions can exhibit markedly different tendencies in pace, repetition, and local exploration, but a one-path design does not expose a mechanism for behavior-specific computation allocation. In other words, representation quality improves, but routing semantics remain implicit.

Mixture-of-experts (MoE) provides a direct mechanism for conditional computation [4, 6, 7, 18]. However, in many recommendation settings, routing is still derived from hidden states. This is practical, but it also entangles two roles that are

better separated: representing sequence content and deciding where computation should go. When both are handled by the same latent signal, expert specialization becomes harder to interpret and debug.

This limitation becomes more serious when the goal is not only accuracy but also controllability and diagnosis. If routing is purely latent, it is difficult to answer basic questions such as: Which behavior pattern activated which expert? Are similar sessions routed consistently? Is an expert specializing in a meaningful regime or merely in an embedding artifact? These questions are central for robust deployment and scientific understanding, but are weakly supported by hidden-state-only routing.

We therefore adopt a simple principle: **decouple routing signals from representation signals**. In RouteRec, hidden states remain the content channel processed by experts, while routing is controlled by compact behavior cues traceable to logs. This separation preserves backbone strength while making routing decisions easier to inspect and reason about.

RouteRec also aligns routing with temporal scope. Behavior evidence near the prediction point is not equivalent to long-range history evidence; forcing a single routing bias for all horizons is unnecessarily rigid. We therefore use a multi-stage design with macro, mid, and micro routing, allowing coarse-to-fine expert allocation that reflects cross-session tendency, within-session evolution, and immediate transitions.

Our goal is not to maximize notation complexity or introduce heavy side branches. Instead, we emphasize a compact, transferable routing view built from broadly available signals. This makes the approach practical across datasets and improves method readability, while still enabling richer conditional computation than shared-path baselines.

This paper also introduces a dedicated *Preliminary* section before the main method. The goal is pragmatic: define the problem statement, notation, and MoE basics once, so the method section can focus on RouteRec-specific design choices.

The main contributions of this work are as follows:

- We identify a core limitation of hidden-state-only routing in sequential MoE recommendation and formalize routing-representation decoupling as a practical design principle.
- We introduce RouteRec, a behavior-guided multi-stage MoE framework that routes with compact cues from standard interaction logs instead of hidden-state-only gating.
- We provide a clear preliminary formulation (problem statement and MoE background) to support readable method development without excessive symbol overhead.
- We show that RouteRec improves recommendation quality over strong baselines and yields more interpretable, behavior-consistent routing patterns in analysis.

2 Related Work

2.1 Sequential Recommendation Backbones

Sequential recommendation has progressed from recurrent and convolutional models to graph and Transformer architectures [5, 8, 20, 22, 26]. Recent methods further strengthen performance with time-aware attention, contrastive objectives, and structural inductive bias [3, 11, 14, 17, 19, 23, 27]. These advances improve representation quality, but they generally retain shared computation across behaviorally different cases.

2.2 Heterogeneous User State Modeling

Many studies explicitly acknowledge non-stationary behavior, including short-term intent modeling and multi-interest extraction [2, 9, 10, 13, 15, 29]. Most approaches handle heterogeneity inside representation learning, for example by attention reweighting or interest decomposition. RouteRec is complementary: it treats heterogeneity as a routing problem and uses explicit behavior cues to modulate computation paths.

2.3 Feature-Aware Sequential Recommendation

Feature-aware methods use timestamps, side information, and feature channels to strengthen sequence encoding [3, 11, 28, 31]. In most cases, these signals are fused into hidden representations. RouteRec uses a different role assignment: features are primarily routing controls, while representation transformation remains expert-specific over hidden states.

2.4 MoE in Recommendation

Classical and modern MoE work demonstrates the value of conditional computation [4, 6, 7, 18]. Recommendation-oriented expert architectures also show strong capacity benefits [1, 12, 16, 21, 30]. Our distinction is not simply adding experts, but making routing semantics behavior-grounded and interpretable through explicit cue design.

3 Preliminary

3.1 Problem Statement

For each user, we observe interactions ordered by time and train the model for next-item prediction. In one training instance, we denote the observed prefix as $x_{1:t}$ and the target as x_{t+1}. Each interaction includes at least an item identity and timestamp, and may additionally include coarse category information depending on the dataset.

To keep the formulation readable, we use two signal channels throughout this paper. The first is the representation channel, where a sequential encoder maps observed prefixes to hidden states $\mathbf{h}_t$. The second is the behavior channel, where lightweight statistics from the same prefix are summarized into a compact cue vector $\mathbf{b}_t$. RouteRec uses $\mathbf{h}_t$ for expert transformation and $\mathbf{b}_t$ for routing decisions. This separation

is the central modeling choice that the later method section instantiates stage by stage.

The behavior channel is intentionally simple: cues are derived from standard log attributes rather than from a deep side encoder. This keeps routing signals stable across datasets and easier to interpret during analysis. In practice, cues capture aspects such as pace, repetition tendency, focus concentration, and popularity exposure. We do not force a large handcrafted taxonomy in this section; instead, we keep the abstraction compact and defer stage-specific feature construction details to Method.

3.2 MoE Background for Sequential Models

In a standard MoE layer with E experts, an expert-weighted output at position t is

$$\tilde{\mathbf{h}}_t = \sum_{e=1}^{E} \pi_{t,e} \operatorname{Expert}_e(\mathbf{h}_t), \tag{1}$$

where $\boldsymbol{\pi}_t \in \mathbb{R}^E$ is a routing distribution with $\sum_e \pi_{t,e} = 1$.

In many practical systems, $\boldsymbol{\pi}_t$ is computed from hidden states. This is effective but often semantically opaque, because routing and content encoding are entangled in the same latent space. RouteRec follows a different view: routing is computed from behavior cues, while hidden states are dedicated to expert transformation. The distinction is conceptually small but operationally important for interpretability and diagnosis.

Another practical point is temporal scope. Sequential recommendation combines long-range tendency and short-range fluctuation, so a single routing granularity can be suboptimal. RouteRec addresses this by applying the same MoE interface at multiple stages with different evidence windows. This keeps notation light while allowing routing behavior to adapt across horizons.

In the final A12 instantiation used in our main experiments, this multi-stage principle is implemented with one shared router composition across all stages, and stage differences are introduced only through temporal evidence windows and routing granularity. The next section formalizes this implementation-faithful view.

4 Method

4.1 RouteRec Overview

Figure 2 summarizes RouteRec. We retain a strong sequential backbone and replace shared feed-forward computation with stage-wise MoE routing. The core interface follows the preliminary setup: hidden states are transformed by experts, and routing is controlled by behavior cues.

Let $\mathcal{T} = \{\text{macro}, \text{mid}, \text{micro}\}$ be the stage set. At stage $s \in \mathcal{T}$, routing uses a stage-specific behavior feature $\mathbf{f}_t^{(s)}$, while experts consume the shared hidden state $\mathbf{h}_t$:

$$\boldsymbol{\pi}_t^{(s)} = \operatorname{softmax}\Big(\operatorname{Router}_s(\mathbf{f}_t^{(s)})/\tau_s\Big), \tag{2}$$

$$\tilde{\mathbf{h}}_t^{(s)} = \sum_{e=1}^{E} \pi_{t,e}^{(s)} \operatorname{Expert}_e^{(s)}(\mathbf{h}_t). \tag{3}$$

Eq. (3) is the common stage interface. In our final A12 model, all stages instantiate this interface with the same wrapper composition, while temporal differences arise from stage windows and routing granularity.

4.2 Behavioral Features from Raw Signals

RouteRec is designed for realistic data availability. It uses signals that are commonly present in interaction logs: item identity, timestamp, optional category, and optional corpus-level popularity. We intentionally avoid heavy side-information encoders in the routing branch. Instead, raw signals are converted into compact scalar summaries that are easy to compute, stable across datasets, and interpretable at analysis time.

For session s_m and target position t, the stage windows are

$$\mathcal{R}_t^{\text{macro}} = [s_{\max(1,m-H)}, \ldots, s_{m-1}], \tag{4}$$

$$\mathcal{R}_t^{\text{mid}} = [x_1, \ldots, x_{t-1}], \tag{5}$$

$$\mathcal{R}_t^{\text{micro}} = [x_{\max(1,t-w)}, \ldots, x_{t-1}], \tag{6}$$

where $H = 5$ and $w = 5$ in our main setting. A shared summary operator $\mathcal{A}(\cdot)$ maps each window to a low-dimensional feature vector:

$$\mathbf{f}_t^{(s)} = \mathcal{A}(\mathcal{R}_t^{(s)}), \qquad s \in \mathcal{T}.$$

The summary dimensions are grouped into four semantic families: *Tempo*, *Focus*, *Memory*, and *Exposure*. Tempo captures pace and interval statistics. Focus captures concentration and switching behavior over coarse categories or item groups. Memory captures recurrence, repetition, and overlap with recent context. Exposure captures interactions with popularity strata and their local drift. This decomposition is semantic rather than architectural: it gives routing modules an interpretable behavioral basis.

Importantly, the feature template is shared across stages, but each stage reads it through a different temporal lens. Macro features reflect persistent tendency and broad preference shape. Mid features track evolving intent inside the current session. Micro features emphasize immediate cues around the prediction target. This shared-template, stage-specific-view design keeps the feature branch compact while preserving temporal specificity.

4.3 Unified A12 Router Composition

The final A12 architecture uses one shared wrapper form at every stage: `w5_exd` (scalar-group conditional product). Experts are partitioned into G groups with C experts per group, so $E = GC$. For stage s and position t, the router

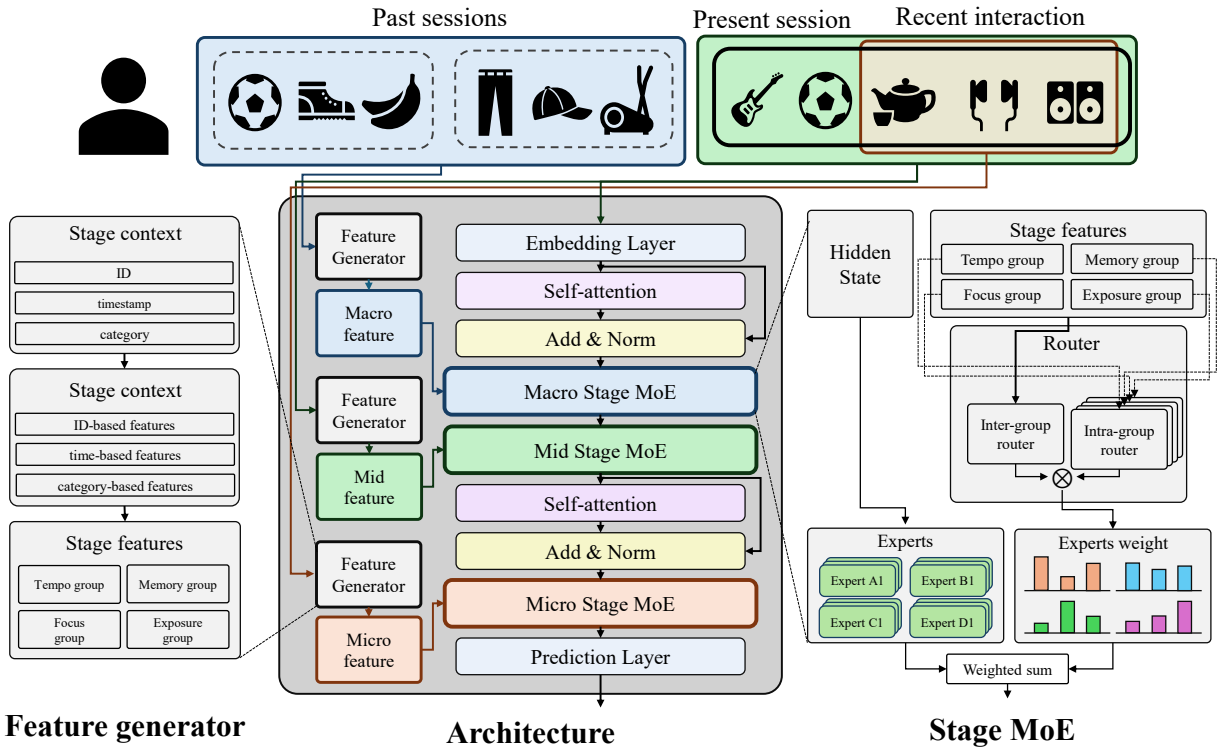

Figure 2: Overall RouteRec framework. (a) RouteRec keeps a strong sequential backbone and replaces shared feed-forward blocks with behavior-guided MoE stages. (b) In the final A12 configuration, all stages use the same scalar-group conditional router composition (`w5_exd`): a group-scalar distribution and a conditional intra-group distribution are multiplied to form expert probabilities. (c) Macro, mid, and micro differ by temporal evidence scope and routing granularity, not by router formula.

produces

$$p_{t,g}^{(s)} = \mathrm{softmax}_g\left(z_{t,g}^{(s)}\right), \tag{7}$$

$$r_{t,g,c}^{(s)} = \mathrm{softmax}_c\left(u_{t,g,c}^{(s)}\right), \tag{8}$$

where $p_{t,g}^{(s)}$ is a group-scalar distribution (E_SCALAR primitive) and $r_{t,g,c}^{(s)}$ is a conditional intra-group distribution (D_COND primitive). The final expert probability is their product:

$$\pi_{t,g,c}^{(s)} = p_{t,g}^{(s)}\, r_{t,g,c}^{(s)}, \tag{9}$$

followed by flattening (g, c) to expert index e. Eq. (9) is the same for macro, mid, and micro.

This unified composition is important for two reasons. First, it gives all stages a shared structural prior, which reduces stage-specific router engineering and simplifies debugging. Second, it keeps diagnostics comparable across stages because differences in routing behavior can be attributed to evidence scope rather than formula changes.

In our implementation, router primitives are driven by behavior cues (feature source), while hidden states remain dedicated to expert transformation in Eq. (3). This preserves the routing-representation decoupling principle introduced in Preliminary.

4.4 Stage Instantiation by Temporal Scope

Although the router formula is shared, stages are not redundant. They differ in temporal evidence and routing granularity.

Macro and mid (session-level routing). Macro and mid compute one routing distribution per sequence from their respective windows, then broadcast it over valid positions inside the sequence. Macro uses broader cross-session history; mid uses within-session prefix evidence. This setting emphasizes stable allocation for slower-changing context.

Micro (token-level routing). Micro computes routing per token near the prediction target using short-horizon context. This enables fast adaptation to immediate transitions without changing the router form itself.

Formally, if Γ_s denotes the granularity operator at stage s, then

$$\boldsymbol{\pi}_t^{(s)} = \Gamma_s\left(\mathcal{R}_t^{(s)}\right), \quad \Gamma_{\mathrm{macro}}, \Gamma_{\mathrm{mid}} : \text{session-level}, \ \Gamma_{\mathrm{micro}} : \text{token-level},$$

Hence, RouteRec's coarse-to-fine behavior comes from stage evidence scope and update frequency, while the underlying router composition remains constant.

4.5 Training Objective

RouteRec is trained with a standard next-item objective and two router-side regularizers: route consistency and z-loss. The first regularizer shapes routing semantics, and the second stabilizes routing numerics.

Let $\mathbf{z}$ be final logits and y be the target item. The main loss is

$$\mathcal{L}_{\mathrm{CE}} = -\log \frac{\exp z_y}{\sum_j \exp z_j}.$$

For route consistency, we first compute a sample-level routing distribution at each stage by averaging token-level routing weights:

$$\bar{\boldsymbol{\pi}}_i^{\,s} = \frac{1}{L_i} \sum_{t=1}^{L_i} \boldsymbol{\pi}_{i,t}^{\,s}$$

where L_i is the valid sequence length. For each stage s, we construct a pair set $\mathcal{P}_s$ from feature-similar samples in the mini-batch and minimize

$$\mathcal{L}_{\mathrm{cons}} = \frac{1}{|\mathcal{T}|} \sum_{s \in \mathcal{T}} \frac{1}{|\mathcal{P}_s|} \sum_{(i,j) \in \mathcal{P}_s} \mathrm{JSD}\big(\bar{\boldsymbol{\pi}}_i^{\,s}, \bar{\boldsymbol{\pi}}_j^{\,s}\big)\,,$$

which encourages neighboring behavioral states to produce compatible routing distributions. In other words, semantically similar inputs should not induce arbitrarily different routing behavior.

To stabilize gating numerics, we adopt z-loss on router logits [33]:

$$\mathcal{L}_z = \frac{1}{|\mathcal{T}|} \sum_{s \in \mathcal{T}} \frac{1}{N_s} \sum_{(i,t) \in \mathcal{V}_s} \left(\log \sum_{e=1}^{E} \exp q_{i,t,e}^{\,s} \right)^2,$$

where $\mathcal{V}_s$ is the set of valid positions and $N_s = |\mathcal{V}_s|$. This term discourages excessive logit scale and helps prevent unstable, overly sharp routing. The final objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda_{\mathrm{cons}}\, \mathcal{L}_{\mathrm{cons}} + \lambda_z\, \mathcal{L}_z,$$

where λ_{cons} and λ_z are trade-off coefficients. This objective jointly optimizes prediction quality, semantic consistency of routing, and numerical stability of expert selection.

5 Experiments

5.1 Experimental Setup

Our main model follows the final A12 configuration in Section 4: macro-mid-micro stage composition, learned routing at all stages, and a unified `w5_exd` wrapper (scalar-group conditional product) for macro, mid, and micro. Stages differ by temporal windows and routing granularity (macro/mid session-level, micro token-level), with feature-driven primitives and bias-specific routing terms disabled. Training uses next-item cross-entropy with route consistency and z-loss, as defined in Section 4.5. Unless otherwise stated, all reported RouteRec results use this configuration.

We compare RouteRec against representative recurrent, graph-based, Transformer-based, frequency-aware, Mamba-based, and MoE-style sequential recommenders. The primary baseline set includes GRU4Rec, SASRec, TiSASRec, and DuoRec [5, 8, 11, 17], plus BSARec, SIGMA, DIF-SR, and FAME [12, 14, 19, 28]. When budget and space allow, we additionally include Caser, SR-GNN, BERT4Rec, FMLP-Rec, and FEARec [3, 20, 22, 26, 32] to broaden architectural coverage.

For fairness, we prioritize a unified training framework: matched embedding dimension, hidden dimension, optimizer family, batch regime, and early-stopping protocol wherever implementation permits. Data preprocessing is documented with sessionization policy, filtering thresholds, sequence truncation rules, and feature availability for timestamp, category, and popularity signals. Evaluation follows standard top-K ranking metrics, including HR@K, NDCG@K, and MRR@K. We report MRR@20 as the primary summary metric and present other metrics for completeness.

5.2 Main Results

This subsection provides the overall performance comparison between RouteRec and all baselines across target datasets. The main table reports ranking metrics at common cutoffs and highlights RouteRec gains over strong Transformer and MoE references.

We focus analysis on three questions. First, does feature-guided routing improve over a strong shared-backbone baseline such as SASRec under matched training conditions? Second, are gains consistent across datasets with different temporal density and behavioral regularity? Third, does RouteRec remain competitive when compared with recent feature-aware or MoE-style alternatives rather than only classical baselines?

Beyond absolute ranking numbers, we summarize relative improvements against SASRec and the strongest non-RouteRec competitor on each dataset. This framing clarifies whether improvements are robust or concentrated in a narrow subset of settings.

5.3 Ablation and Diagnostic Analysis

This subsection analyzes why RouteRec works through controlled ablations and routing diagnostics. We first evaluate router-design variants (hierarchical, hybrid, direct combinations) to test whether stage-specific routing bias is necessary. We then evaluate feature-family ablations to measure the contribution of Tempo, Focus, Memory, and Exposure cues.

Next, we examine stage composition by removing or altering macro, mid, and micro branches, verifying whether coarse-to-fine routing is materially better than a single-stage or uniformly structured alternative. We also include corrupted-feature sanity checks to confirm that gains are tied to meaningful behavioral cues rather than incidental regularization effects.

Finally, we present routing diagnostics, including expert usage distribution, stage-wise route entropy, and consistency

behavior under similar feature states. These analyses connect quantitative improvements to the intended mechanism: behavior-aware specialization rather than opaque capacity inflation.

6 Conclusion

The conclusion will restate the central claim of the paper: compact behavioral cues can serve as explicit routing guidance for mixture-of-experts sequential recommendation. It will summarize the empirical findings and discuss portability, interpretability, and future extensions of the framework.

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